COMICCRAFT — PERFORMANCE TESTING

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TEST CASE 1: Standard Comic Generation

Input: Prompt="A brave fox exploring an enchanted forest",

Character="Fox", Setting="forest", Tone="Dramatic",

Style="Anime"

Expected: 5 panels with images and narration

Result: PASSED

Time: ~45 seconds

TEST CASE 2: Funny Comic in City

Input: Prompt="A cat tries to open a restaurant",

Character="Whiskers", Setting="city", Tone="Funny",

Style="Comic Book"

Expected: 5 panels with humorous narration

Result: PASSED

Time: ~42 seconds

TEST CASE 3: Space Adventure

Input: Prompt="A robot astronaut discovers a new planet",

Character="Robo", Setting="space", Tone="Mysterious",

Style="Realistic"

Expected: 5 panels with sci-fi themed illustrations

Result: PASSED

Time: ~50 seconds

TEST CASE 4: Educational Comic

Input: Prompt="A young girl learns about photosynthesis",

Character="Maya", Setting="school", Tone="Light-hearted",

Style="Watercolor"

Expected: 5 panels explaining the concept

Result: PASSED

Time: ~48 seconds

TEST CASE 5: Image Generation Reliability

Action: Run 10 consecutive comic generations (50 images total)

Result: 48 out of 50 images generated successfully

2 images failed → showed "Image unavailable" placeholder

Status: PASSED (fallback worked correctly)

TEST CASE 6: Empty Prompt Handling

Action: Submit form with empty prompt

Result: HTML validation prevents submission (required field)

Status: PASSED

TEST CASE 7: Gemini Rate Limit

Action: Generate 20 comics rapidly

Result: 429 RESOURCE_EXHAUSTED error shown in preview, no crash

Status: PASSED

TEST CASE 8: Invalid JSON from Gemini

Action: Occasionally the model returns extra text around JSON

Result: Regex extraction recovers the JSON array correctly

Status: PASSED

TEST CASE 9: Panel Splitting Fallback

Action: Generate comic where Gemini doesn't use "**Panel X" markers

Result: Fallback even-split algorithm produces 5 balanced chunks

Status: PASSED

TEST CASE 10: Preview Rendering

Action: Load preview page with all 5 panels

Result: All images, titles, descriptions, and narration display correctly

Status: PASSED

PERFORMANCE METRICS

- Average outline generation: 5 seconds
- Average story generation: 8 seconds
- Average per-image generation: 6 seconds
- Average total time (5 panels): 45 seconds
- Memory usage: under 200 MB
- Browser rendering time: under 1 second

CONCLUSION

All 10 test cases passed. The application performs reliably within acceptable limits for a free-tier AI application.